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SEAT SWITCH DEVICE AND SEAT INCLUDING THE SAME

Abstract

A seat switch device capable of preventing unintentional operation of a seat, and a seat including the same, are provided.
A seat switch device **10** including a base portion **22** and an operation knob **30** for operating a seat **90** (see FIG. **4**). The operation knob **30** is provided at a device front side with respect to the base portion **22.** The seat switch device **10** also includes a guard member **50** that is provided at the device front side with respect to the base portion **22.** The guard member **50** functions so as to guard the operation knob **30.**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-033245, filed on Mar. 5, 2024, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a seat switch device and to a seat including the same.

Related Art

[0003] A seat switch device (slide-switch structural body) including an operation knob (slide knob and reclining knob) is disclosed in Japanese Patent Application Laid-Open (JP-A) No. 2010-251028.

SUMMARY

[0004] In the seat switch device configured as described above, there is a possibility that a seat might be operated unintentionally were some sort of object to hit the operation knob.

[0005] An object of the present disclosure is to provide a seat switch device capable of preventing operation of a seat unintentionally, and a seat provided with the same.

[0006] A seat switch device according to a first aspect includes a base portion, an operation knob that operates a seat and that is provided at a device front side with respect to the base portion, and a guard member that is provided at the device front side with respect to the base portion and that guards the operation knob.

[0007] In the first aspect, the seat switch device includes the base portion and the operation knob that operates the seat. The operation knob is provided at the device front side with respect to the base portion.

[0008] This thereby enables the base portion to be arranged at the inside of the exterior member, and the operation knob provided at the device front side with respect to the base portion to be arranged at the outside of an exterior member. The seat can then be operated by the hand of an occupant touching the operation knob and moving the operation knob.

[0009] However, with a seat switch device merely configured as described above, there would be a possibility that a seat might be operated unintentionally were some sort of object to hit the operation knob.

[0010] Thus in the first aspect, the seat switch device includes the guard member provided at the device front side with respect to the base portion. The guard member functions so as to guard the operation knob.

[0011] Thus due to being able to dispose the guard member at the outside of the exterior member in the same way as the operation knob, objects hitting the operation knob can be prevented by the guard member. Unintentional operation of the seat can be prevented as a result.

[0012] Note that in exemplary embodiments described below the seat switch device is arranged with a device front side facing toward a vehicle width direction outside. However, the present aspect is not limited thereto. For example, the device front side may be aligned with a vehicle front side, or may be aligned with a vehicle rear side.

[0013] Note that in exemplary embodiments described below the guard member is fixed so as to be immovable with respect to the base portion. This thereby enables confusion of the occupant between the guard member and the operation knob to be prevented. However, the present aspect is not limited thereto, and the guard member may be configured so as to move with respect to the base portion.

[0014] Note that in the exemplary embodiments described below an end on the device front side of

the guard member is positioned further toward the device front side than an end on the device front side of the operation knob. However, the present aspect is not limited thereto.
[0015] Note that in the exemplary embodiments described below the guard member includes a covering portion that covers the operation knob from the device front side. However, the present aspect is not limited thereto.
[0016] A seat switch device according to a second aspect is the first aspect, wherein an end at a device front side of the guard member is positioned further toward the device front side than an end at a device front side of the operation knob.
[0017] In the second aspect, the end at the device front side of the guard member is positioned further toward the device front side than the end at the device front side of the operation knob.
[0018] This thereby enables objects to be effectively prevented from hitting the operation knob.
[0019] Note that in exemplary embodiments described below the guard member includes a covering portion that covers the operation knob from the device front side. However, the present aspect is not limited thereto.
[0020] A seat switch device according to a third aspect is the first or the second aspect, wherein the guard member includes a covering portion that covers the operation knob from the device front side.
[0021] In the third aspect, the guard member includes the covering portion that covers the operation knob from the device front side.
[0022] This thereby enables objects to be effectively prevented from hitting the operation knob from the device front side by the covering portion.
[0023] A seat switch device according to a fourth aspect is any of the first aspect to the third aspect, wherein the guard member includes an outer face portion configuring an outer face of the guard member, and the outer face portion includes a top face portion configuring a top face facing toward the device front side with the top face portion covering the operation knob from the device front side, and a side face portion configuring a side face facing in a direction perpendicular to a device front direction.
[0024] In the fourth aspect, the guard member includes the outer face portion configuring the outer face of the guard member. The outer face portion includes the top face portion configuring the top face facing toward the device front side, and the side face portion configuring the side face facing in the direction perpendicular to the device front direction.
[0025] The durability of the guard member is accordingly readily secured while securing good styling.
[0026] Moreover, in the fourth aspect the top face portion covers the operation knob from the device front side.
[0027] This thereby enables the top face portion of the guard member to function as a covering portion covering the operation knob from the device front side.
[0028] A seat switch device according to a fifth aspect is the fourth aspect, wherein the side face portion includes an opening exposing a portion of the operation knob.
[0029] In a fifth aspect, the side face portion includes the opening exposing a portion of the operation knob.
[0030] The side faces of the guard member are accordingly positioned on each side of the opening exposing a portion of the operation knob. As a result thereof, the guard member can effectively function as an operation guide for operating the operation knob.
[0031] Note that in the exemplary embodiments described below, a portion of the operation knob is visible through the opening in the guard member when viewed from the device front side. This accordingly facilitates operation of the operation knob. However, the present aspect is not limited thereto, and the entire operation knob may be hidden when viewed from the device front face.
[0032] A seat switch device according to a sixth aspect is any one of the first aspect to the fifth aspect, wherein the guard member includes a covering portion that covers the operation knob from

the device front side, and two pillar portions that support the covering portion from a device rear side with the two pillar portions respectively disposed on each side of the operation knob.
[0033] In the sixth aspect, the guard member includes the covering portion that covers the operation knob from the device front side and the two pillar portions that support the covering portion from the device rear side. The two pillar portions are disposed on each side of the operation knob.
[0034] When, for example, load toward the device rear side has acted on the guard member, the covering portion can be appropriately supported by the two pillar portions. The durability of the guard member is readily secured as a result.
[0035] A seat switch device according to a seventh aspect is any one of the first aspect to the sixth aspect, wherein the seat switch device includes a guard projection portion that projects from the base portion toward the device front side, and the pillar portions include a side face portion configuring a side face facing in a direction perpendicular to a device front direction, and a fitting portion that fits together with the guard projection portion.
[0036] In the seventh aspect, the seat switch device includes the guard projection portion that projects from the base portion toward the device front side. The pillar portions include the side face portion configuring the side face facing in the direction perpendicular to the device front direction, and the fitting portion that fits together with the guard projection portion.
[0037] The covering portion can accordingly be more effectively supported than cases in which, for example, the pillar portions are not provided with the fitting portion. The durability of the guard member is accordingly readily secured as a result.
[0038] A seat switch device according to an eighth aspect is any one of the first aspect to the seventh aspect, wherein the seat switch device is arranged with a device front side facing toward a seat width direction out side, the operation knob operates a seatback, the guard member has a shape with a length direction substantially along a seat height direction, the guard member includes side face portions configuring side faces of the guard member, and among the side faces, a rear side face facing toward a seat rear side extends in a direction inclined toward a seat rear on progression in a seat upward direction.
[0039] In the eighth aspect, the seat switch device is arranged with the device front side facing toward the seat width direction outside, and the operation knob is an operation knob for operating the seatback. The guard member has a shape with a length direction substantially along the seat height direction.
[0040] An occupant is accordingly able, by touching the guard member and gripping the profile thereof (the shape with a length direction substantially along the seat height direction), to recognize that this is the guard member corresponding to the operation knob for operating the seatback.
[0041] Moreover, in the eighth aspect the guard member includes the side face portion configuring the side face of the guard member. From out of the side faces, the rear side face facing toward the seat rear side extends in the direction inclined toward the seat rear on progression in the seat upward direction.
[0042] Any finger other than the thumb of the occupant is accordingly easily placed along the side face of the guard member facing the rear side. The operability can be improved as a result thereof.
[0043] Note that in the exemplary embodiments described below a side face of the guard member facing the front side also extends in a direction inclined toward the seat rear on progression in the seat upward direction. However, the present aspect is not limited thereto.
[0044] Note that “substantially along the seat height direction” is a definition encompassing the seat height direction, and means a direction inclined in a range of ±40° with respect to the seat height direction.
[0045] A seat switch device according to a ninth aspect is any one of the first aspect to the seventh aspect, wherein the seat switch device is arranged with a device front side facing toward a seat width direction out side, the operation knob operates a seat cushion, the guard member has a shape

with a length direction along a seat front-rear direction, the guard member includes side face portions configuring side faces of the guard member, and among the side faces, a lower side face facing in a seat downward direction extends in the seat front-rear direction.
[0046] In the ninth aspect, the seat switch device is arranged with the device front side facing toward the seat width direction outside, and the operation knob is the operation knob for operating the seat cushion. The guard member has a shape with the length direction along the seat front-rear direction.
[0047] Thus by touching the guard member and gripping the profile thereof (the shape with a length direction along the seat front-rear direction), the occupant is able to recognize that this is the guard member corresponding to the operation knob for operating the seat cushion.
[0048] Moreover, in the ninth aspect, the guard member includes the side face portions configuring the side faces of the guard member. From out of the side faces, the lower side face facing toward the seat downward side extends along the seat front-rear direction.
[0049] This means that any finger other than the thumb of the occupant is easily placed along the lower side face of the guard member. The operability can be improved as a result.
[0050] Note that in the exemplary embodiments described below, from out of the side faces of the guard member, the upper side face facing the seat upper side extends also along the seat front-rear direction. However, the present aspect is not limited thereto.
[0051] A seat according to a tenth aspect includes the seat switch device according to any one of the first aspect to the ninth aspect.
[0052] In the tenth aspect, the seat includes the seat switch device described above.
[0053] A seat can accordingly be realized that is capable of preventing unintentional operation of the seat.
[0054] As described above, the present disclosure provides a seat switch device capable of preventing unintentional operation of a seat, and a seat including the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0055] Exemplary embodiments of the present disclosure will be described in detail based on the following figures, wherein:
[0056] FIG. **1** is an exploded perspective diagram of a seat switch device of an exemplary embodiment;
[0057] FIG. **2** is a front view of part of a seat switch device of an exemplary embodiment as viewed from in front of the device;
[0058] FIG. **3** is an expanded perspective view of a first operation knob; and
[0059] FIG. **4** is a schematic side view of a seat including a seat switch device.
DETAILED DESCRIPTION
[0060] Description follows regarding a seat switch device **10** according to an exemplary embodiment of the present disclosure, and a seat **90** provided with the same.
[0061] Note that in the drawings, as appropriate, arrow FR indicates a vehicle front direction, arrow UP indicates a vehicle upward direction, and arrow OUT indicates a vehicle width direction outside. Moreover, in the present exemplary embodiment, the device front side of the seat switch device **10** is aligned with the vehicle width direction outside.
[0062] FIG. **1** is an exploded perspective view of the seat switch device **10**, and FIG. **2** is a front view of part of the seat switch device **10**, as viewed from in front of the device. Note that FIG. **2** schematically depicts parts hidden by other members using broken lines.
[0063] As illustrated in FIG. **1**, the seat switch device **10** includes a main body **20**, a first operation knob **30**A, a second operation knob **30**B, a first guard member **50**A, and a second guard member

50B. The first operation knob **30**A and the second operation knob **30**B will simply be referred to as operation knob **30** when not particularly distinguishing therebetween. The first guard member **50**A and the second guard member **50**B will also simply be referred to as guard member **50** when not particularly distinguishing therebetween.

[0064] The operation knob **30** is a knob for operating the seat **90** (see FIG. **4**). Specifically, the first operation knob **30**A is an operation knob for operating a seat cushion **92**. The second operation knob **30**B is an operation knob for operating a seatback **94**.

[0065] Operations of the seat cushion **92** include a slide operation, a vertical operation, and a lift operation. The slide operation means an operation to move of the seat cushion **92** in the front-rear direction. The vertical operation means an operation to move a front end portion of the seat cushion **92** in the height direction. A lift operation means an operation to move the entire seat cushion **92** in the height direction. Operation of the seatback **94** includes a reclining operation.

[0066] The guard members **50** are members to guard the operation knobs **30**. In other words, the guard members **50** function so as to prevent operation of the operation knobs **30** were an object to unintentionally hit the operation knobs **30**. The first guard member **50**A corresponds to the first operation knob **30**A, and the second guard member **50**B corresponds to the second operation knob **30**B.

[0067] The main body **20** includes a base portion **22**, a plurality of operation projection portions **24**A, **24**B projecting from the base portion **22** toward the device front side, and a plurality of guard projection portions **26**A**1**, **26**A**2**, **26**A**3**, **26**B that project from the base portion **22** toward the device front side.

[0068] The base portion **22** includes a case **23**, and in-built components (non-illustrated) are housed in the case **23**.

[0069] The plural operation projection portions **24**A, **24**B include a plurality of (two) first operation projection portions **24**A and a single second operation projection portion **24**B. These will simply be referred to as operation projection portions **24** when not particularly distinguishing therebetween.

[0070] The first operation knob **30**A is attached to the plural first operation projection portions **24**A. The second operation knob **30**B is attached to the second operation projection portion **24**B.

[0071] The operation projection portions **24** are moved by an occupant moving the operation knobs **30**, with contact states inside the case **23** being switched thereby. This thereby enables the seat **90**, which is a power seat, to be operated.

[0072] The plural guard projection portions **26**A**1**, **26**A**2**, **26**A**3**, **26**B including plural (six) first guard projection portions **26**A**1**, **26**A**2**, **26**A**3**, and plural (two) second guard projection portions **26**B. These will simply be referred to as guard projection portions **26** when not particularly distinguishing therebetween.

[0073] The first guard member **50**A is attached to the plural first guard projection portions **26**A**1**, **26**A**2**, **26**A**3**. The second guard member **50**B is attached to the plural second guard projection portions **26**B.

[0074] The guard projection portions **26** are fixed so as to be immovable with respect to the base portion **22**. This means that the guard members **50** are also fixed immovable with respect to the base portion **22**.

[0075] Note that a vehicle seat exterior member **96** (seat side shield, see FIG. **4**) is arranged at the device front side of the base portion **22**. Holes are formed in the seat side shield for the projection portions **24**, **26** to respectively pass through, with each of the projection portions **24**, **26** being in a state projecting through these holes to outside the vehicle seat exterior member **96**. The operation knob **30** and the guard member **50** are attached to each of the projection portions **24**, **26** in this state.

[0076] Next, detailed description follows regarding configuration related to the first guard member **50**A.

[0077] The two first operation projection portions **24**A include a first operation projection portion **24**A at a vehicle front side and a first operation projection portion **24**A a vehicle rear side.
[0078] The six first guard projection portions **26**A**1**, **26**A**2**, **26**A**3** include two central guard projection portions **26**A**1**, and four outside guard projection portions **26**A**2**, **26**A**3**.
[0079] The central guard projection portions **26**A**1** are attached to a front-rear direction center portion of the first guard member **50**A. The front two outside guard projection portions **26**A**2**, **26**A**3** are attached to a front end portion of the first guard member **50**A, and the rear two outside guard projection portions **26**A**2**, **26**A**3** are attached to a rear end portion of the guard member **50**.
[0080] The central guard projection portions **26**A**1** each have a substantially circular pillar shape with an axis along the device front-rear direction. The outside guard projection portions **26**A**2**, **26**A**3** each have a plate shape with a plate thickness directions facing in the device height direction. The height dimension of the central guard projection portions **26**A**1** is larger than the height dimension of the outside guard projection portions **26**A**2**, **26**A**3**.
[0081] The four outside guard projection portions **26**A**2**, **26**A**3** include two upper guard projection portions **26**A**2** and two lower guard projection portions **26**A**3**. The width dimension (seat front-rear direction dimension) of the upper guard projection portions **26**A**2** is larger than the width direction of the lower guard projection portions **26**A**3**.
[0082] As illustrated in FIG. **2**, the first operation knob **30**A includes a pair of vertical exposed portions **31**K**1**, **31**K**2**, a pair of lift exposed portions **32**K**1**, **32**K**2**, and a pair of slide exposed portions **31**L**1**, **32**L**1**.
[0083] The occupant is able to perform a vertical operation on the seat cushion **92** by pressing one or other of the pair of vertical exposed portions **31**K**1**, **31**K**2**.
[0084] The occupant is able to perform a lift operation on the seat cushion **92** by pressing one or other of the pair of lift exposed portions **32**K**1**, **32**K**2**.
[0085] The occupant is able to perform a slide operation on the seat cushion **92** by pressing one or other of the pair of slide exposed portions **31**L**1**, **32**L**1**.
[0086] As illustrated in FIG. **3**, the first operation knob **30**A includes a front portion **31**, a rear portion **32**, and a coupling portion **33** coupling the front portion **31** and the rear portion **32** together in the front-rear direction.
[0087] The pair of vertical exposed portions **31**K**1**, **31**K**2** and the front slide exposed portion **31**L**1** are formed to the front portion **31**. The pair of lift exposed portions **32**K**1**, **32**K**2** and the rear slide exposed portion **32**L**1** are formed to the rear portion **32**. The front portion **31** is attached to the first operation projection portion **24**A at the front side from out of the two first operation projection portions **24**A. The rear portion **32** is attached to the first operation projection portion **24**A at the rear side from out of the two first operation projection portions **24**A.
[0088] The front portion **31** includes an up-down extension portion **31**K extending in the height direction, and a forward extension portion **31**L extending in a forward direction from the up-down extension portion **31**K. The forward extension portion **31**L is arranged between the two outside guard projection portions **26**A**2**, **26**A**3** on the front side.
[0089] The rear portion **32** includes an up-down extension portion **31**K extending in the height direction, and a rear extension portion **32**L extending in a rearward direction from the up-down extension portion **32**K. The rear extension portion **32**L is arranged between the two outside guard projection portions **26**A**2**, **26**A**3** at the rear side.
[0090] The coupling portion **33** is arranged between the two central guard projection portions **26**A**1**.
[0091] The first guard member **50**A includes an outer face portion **51** configuring an outer face of the first guard member **50**A, and plural fitting portions **52** (see FIG. **2**) that fit together with the plural first guard projection portions **26**A**1**, **26**A**2**, **26**A**3**. The fitting portions **52** are provided at the inside of the outer face portion **51** so as not to be visible to the occupant.
[0092] The outer face portion **51** includes a top face portion **53** configuring a top face facing the

device front side, and side face portions **54** configuring side faces facing in directions perpendicular to the device front direction. The outer face portion **51** is configured such that the top face and the side faces are connected together by smooth curved faces. The top face portion **53** of the first guard member **50**A is configured to cover the first operation knob **30**A from the device front side. In the following the top face portion **53** of the first guard member **50**A is sometimes referred to as a covering portion **53**.
[0093] The side face portions **54** of the first guard member **50**A include plural (six) openings **55** that expose the exposed portions **31**K**1**, **31**K**2**, **32**K**1**, **32**K**2**, **31**L**1**, **32**L**1** of the first operation knob **30**A. The plural openings **55** are provided in a number corresponding to the plural number of the exposed portions **31**K**1**, **31**K**2**, **32**K**1**, **32**K**2**, **31**L**1**, **32**L**1**. The side face portions **54** are separated into plural (six) portions **54** by the plural openings **55**. In the following the respective separated portions **54** are sometimes referred to as side face portions **54**.
[0094] As illustrated in FIG. **2**, from out of the six separated side face portions **54**, four side face portions **54** are positioned at the four corners of the first guard member **50**A formed in a substantially quadrangular shape (specifically a trapezoidal shape) when looking at the device front face. The side faces where the four side face portions **54** are formed at the four corner positions are configured by smooth curved faces.
[0095] Next, detailed description follows regarding configuration related to the second guard member **50**B. Note that the same reference numerals will be appended in the drawings to similar configuration to the configuration related to the first guard member **50**A.
[0096] As illustrated in FIG. **1**, the plural second guard projection portions **26**B include an upper second guard projection portion **26**B and a lower second guard projection portion **26**B. The upper second guard projection portion **26**B is positioned further to the seat rear side than the lower second guard projection portion **26**B.
[0097] As illustrated in FIG. **2**, the second operation knob **30**B includes a pair of reclining exposed portions **31**M**1**, **31**M**2**.
[0098] The occupant is able to perform a reclining operation on the seatback **94** by pressing one or other of the pair of reclining exposed portions **31**M**1**, **31**M**2**.
[0099] The second operation knob **30**B is attached to the second operation projection portions **24**B.
[0100] The second operation knob **30**B has a shape that extends in substantially the seat front-rear direction. A front end portion of the second operation knob **30**B is configured by the front reclining exposed portion **31**M**1**, and a rear end portion of the second operation knob **30**B is configured by the rear reclining exposed portion **31**M**2**. The second operation knob **30**B is arranged between an upper second guard projection portion **26**B and a lower second guard projection portion **26**B.
[0101] As illustrated in FIG. **2**, the second guard member **50**B has a shape with a length direction substantially along the seat height direction. Specifically, the second guard member **50**B has a shape with a length direction along a direction inclined slightly toward the seat rear side on progression in the seat upward direction.
[0102] The second guard member **50**B includes an outer face portion **51** configuring an outer face of the second guard member **50**B, and two fitting portions **52** that fit together with the two second guard projection portions **26**B. The fitting portions **52** are provided at the inside of the outer face portion **51** so as not to be visible to the occupant.
[0103] The outer face portion **51** includes a top face portion **53** configuring a top face facing the device front side, and side face portions **54** configuring side faces facing in directions perpendicular to the device front direction. The outer face portion **51** is configured such that the top face and the side faces are connected together by smooth curved faces. The top face portion **53** of the second guard member **50**B is configured to cover the second operation knob **30**B from the device front side. In the following the top face portion **53** of the second guard member **50**B is sometimes referred to as a covering portion **53**.
[0104] The side face portions **54** of the second guard member **50**B include two openings **55** that

expose plural exposed portions **31**M**1**, **31**M**2** of the second operation knob **30**B. The plural openings **55** are provided in a number corresponding to the plural number of the exposed portions **31**M**1**, **31**M**2**. The side face portions **54** are separated into two portions **54** by the two openings **55**. In the following the respective separated portions **54** are sometimes referred to as side face portions **54**.

Operation and Advantageous Effects

[0105] Next, description follows regarding the operation and advantageous effects of the present exemplary embodiment.

[0106] In the present exemplary embodiment, as illustrated in FIG. **1** and FIG. **2**, the seat switch device **10** includes the base portion **22**, and the operation knobs **30** for operating the seat **90** (see FIG. **4**). The operation knobs **30** are provided at the device front side with respect to the base portion **22**.

[0107] This means that base portion **22** can be arranged inside the vehicle seat exterior member **96** (see FIG. **4**), and the operation knobs **30** provided at the device front side with respect to the base portion **22** can be arranged at the outside of the vehicle seat exterior member **96**. The occupant is accordingly able to operate the seat **90** by touching the operation knobs **30** and moving the operation knobs **30**.

[0108] In the seat switch device **10** configured as described above, were some type of object to hit the operation knobs **30**, there would be a possibility that the seat **90** might be operated unintentionally.

[0109] Thus in the present exemplary embodiment, the seat switch device **10** includes the guard members **50** provided at the device front side with respect to the base portion **22**. The guard members **50** function so as to guard the operation knobs **30**.

[0110] This accordingly enables the guard members **50** to, the same as the operation knobs **30**, be arranged at the outside of the vehicle seat exterior member **96**, enabling an object hitting the operation knobs **30** to be prevented by the guard members **50**. As a result thereof unintentional operation of the seat **90** can be prevented.

[0111] Moreover, in the present exemplary embodiment, ends (top face portions **53**) at the device front side of the guard members **50** are positioned further to the device front side than the device front side ends of the operation knobs **30**.

[0112] This thereby enables an object to be effectively prevented from hitting the operation knobs **30**.

[0113] Moreover, in the present exemplary embodiment, the guard members **50** each include the covering portion **53** covering the operation knob **30** from the device front side.

[0114] This thereby enables an object hitting the operation knob **30** from the device front side to be effectively prevented by the covering portion **53**.

[0115] Moreover, in the present exemplary embodiment, the guard members **50** each include the outer face portion **51** configuring the outer face of the guard member **50**. The outer face portion **51** includes the top face portion **53** configuring the top face facing toward the device front side, and the side face portions **54** configuring side faces facing in directions perpendicular to the device front direction.

[0116] This accordingly means that the durability of the guard member **50** is easily secured while also securing good styling.

[0117] In particular, in the present exemplary embodiment, the outer face portion **51** is configured such that the top face and the side faces are connected by smooth curved faces.

[0118] The guard member **50** accordingly secures a pleasant feeling to the touch.

[0119] Moreover, in the present exemplary embodiment, the top face portion **53** covers the operation knob **30** from the device front side.

[0120] This accordingly enables the top face portion **53** of the guard member **50** to function as the covering portion **53** covering the operation knob **30** from the device front side.

[0121] Moreover, in the present exemplary embodiment, the side face portions **54** include the openings **55** that expose portions of the operation knobs **30**.
[0122] This means that side faces of the guard members **50** are positioned on each side of the respective openings **55** that expose portions of the operation knob **30**. As a result thereof, the guard members **50** can effectively function as operation guides for operation of the operation knobs **30**.
[0123] In particular, in the present exemplary embodiment, as illustrated in FIG. **2**, portions of each of the operation knobs **30** are visible from the openings **55** of the guard members **50** when looking from the device front side. This facilitates operation of the operation knobs **30**.
[0124] Moreover, in the present exemplary embodiment, as illustrated in FIG. **2**, the guard members **50** each include the covering portion **53** covering the operation knob **30** from the device front side, and the two pillar portions **52**, **54** supporting the covering portion **53** from the device rear side. The two pillar portions **52**, **54** are arranged on each side of the operation knobs **30**.
[0125] Thus when, for example, load toward the device rear side has acted on the guard member **50**, the covering portion **53** can be appropriately supported by the two pillar portions **52**, **54**. As a result thereof, durability of the guard member **50** is easily secured.
[0126] Moreover, in the present exemplary embodiment, as illustrated in FIG. **1**, the seat switch device **10** includes the guard projection portions **26** projecting toward the device front side from the base portion **22**. Moreover, as illustrated in FIG. **2**, pillar portions **52**, **54** include the side face portions **54** configuring the side faces facing in directions perpendicular to the device front direction, and the fitting portions **52** for fitting together with the guard projection portions **26**.
[0127] This accordingly enables the covering portions **53** to be more effectively supported than cases in which, for example, the pillar portions **52**, **54** do not include fitting portions **52**. The durability of the guard members **50** is easily secured as a result.
[0128] Moreover, in the present exemplary embodiment, as illustrated in FIG. **4**, the seat switch device **10** is arranged with the device front side facing toward the seat width direction outside, and the second operation knob **30**B is the operation knob **30** for operating the seatback **94**. As illustrated in FIG. **2**, the second guard member **50**B has a shape with a length direction substantially along the seat height direction.
[0129] This means that the occupant is able, by touching this guard member **50** and gripping the profile thereof (the shape with a length direction substantially along the seat height direction), to recognize that this is the second guard member **50**B corresponding to the second operation knob **30**B for operating the seatback **94**.
[0130] Moreover as illustrated in FIG. **2**, in the present exemplary embodiment the second guard member **50**B includes the side face portions **54** configuring the side faces of the second guard member **50**B. From out of the side faces of the second guard member **50**B, the rear side face facing the seat rear side extends in direction inclined toward the seat rear on progression in the seat upward direction.
[0131] This means that any finger other than the thumb of the occupant is easily placed along the side face of the guard member **50** facing the rear side. An improvement in operability can be achieved as a result thereof.
[0132] Moreover as illustrated in FIG. **4**, in the present exemplary embodiment, the seat switch device **10** is device disposed with the device front side facing toward the seat width direction outside, and the first operation knob **30**A is the operation knob **30** for operating the seat cushion. The first guard member **50**A accordingly has a shape with a length direction along the seat front-rear direction.
[0133] This means that, by touching this guard member **50** and gripping the profile thereof (the shape with a length direction along the seat front-rear direction), the occupant is able to recognize that this is the first guard member **50**A corresponding to the first operation knob **30**A for operating the seat cushion **92**.
[0134] Moreover as illustrated in FIG. **2**, in the present exemplary embodiment the first guard

member **50**A includes the side face portions **54** configuring the side faces of the first guard member **50**A. From out of the side faces of the first guard member **50**A, the lower side face facing the seat downward side extends along the seat front-rear direction.
[0135] Any finger other than the thumb of the occupant is accordingly easily placed along the lower side face of the first guard member **50**A. An improvement in operability can be achieved as a result thereof.
[0136] Moreover as illustrated in FIG. **4**, in the present exemplary embodiment the seat switch device **10** is arranged with the device front side facing toward the vehicle width direction outside.
[0137] This means that, for example, there might be a concern regarding door trim of a side door hitting the operation knob **30** when a side-impact collision occurred, and regarding the seat **90** being unintentionally operated.
[0138] However, in the present exemplary embodiment, the guard member **50** is provided and so unintentional operation when a side-impact collision occurs can be prevented.
[0139] Moreover, in the present exemplary embodiment, as illustrated in FIG. **2**, the first guard member **50**A has a shape that is an elongated trapezoidal shape, with a longer upper base than lower base as viewed from the device front side.
[0140] This means that the occupant easily grips the first guard member **50**A, and good operability is secured.
[0141] Furthermore, in the present exemplary embodiment, the upper guard projection portions **26**A**2** have width dimensions larger than the lower guard projection portions **26**A**3**.
[0142] This thereby enables the guard projection portions **26** to be made large effectively within limitations to the shape of the first guard member **50**A as described above. The durability of the first guard member **50**A is readily secured as a result thereof.
[0143] Moreover, in the present exemplary embodiment, as illustrated in FIG. **1**, the front portion **31** and the rear portion **32** of the first operation knob **30**A are connected together by the coupling portion **33**.
[0144] This means that the number of components can be reduced compared to cases in which the first operation knob **30**A is separated into the front portion **31** and the rear portion **32**.
[0145] Furthermore, in the present exemplary embodiment the central guard projection portions **26**A**1** are arrange above and below the coupling portion **33**. As illustrated in FIG. **3**, the coupling portion **33** has a height dimension that is smaller than that of either the forward extension portion **31**L or the rear extension portion **32**L.
[0146] This means that the size of the central guard projection portions **26**A**1** is readily secured. The durability of the first guard member **50**A is readily secured as a result thereof. In the present exemplary embodiment, the height dimension of the central guard projection portions **26**A**1** is actually larger than the height dimension of the outside guard projection portions **26**A**2**, **26**A**3**.
[0147] Moreover as illustrated in FIG. **2**, in the present exemplary embodiment the radius of curvature at a boundary position between the rear side face and the lower side face of the second guard member **50**B is greater than the radius of curvature at a boundary position between the front side face and the lower side face.
[0148] This means that the radius of curvature is large at a position where the ball of the finger of the occupant is liable to be positioned, raising operability.
[0149] Moreover as illustrated in FIG. **2**, in the present exemplary embodiment, the radius of curvature at a boundary position between the front side face and the upper side face of the second guard member **50**B is larger than the radius of curvature at a boundary position between the rear side face and the upper side face.
[0150] This means that the radius of curvature is large at the position where the thenar eminence of the occupant is liable to be positioned, and operability is raised.
Supplementary Explanation to Above Exemplary Embodiments
[0151] Note that in the above exemplary embodiments the first guard member **50**A and the second

guard member **50**B are provided so as to respectively correspond to the first operation knob **30**A and the second operation knob **30**B. However, the present disclosure is not limited thereto, and a guard member corresponding to only one of the operation knobs may be provided alone.
[0152] Moreover, the direction of movement of the operation knob **30** in the above exemplary embodiment is a direction perpendicular to the device front-rear direction. However, the present disclosure is not limited thereto.
[0153] Moreover, in the above exemplary embodiments the seat switch device **10** includes the guard projection portions **26** projecting from the base portion **22** toward the device front side. However, the present disclosure is not limited thereto.
EXPLANATION OF REFERENCE NUMERALS
[0154] **10** seat switch device [0155] **22** base portion [0156] **26**A**1**, **26**A**2**, **26**A**3** first guard projection portion (guard projection portion) [0157] **26**B second guard projection portion (guard projection portion) [0158] **30**A first operation knob (operation knob) [0159] **30**B second operation knob (operation knob) [0160] **50**A first guard member (guard member) [0161] **50**B second guard member (guard member) [0162] **51** outer face portion [0163] **52** fitting portion (pillar portion) [0164] **53** top face portion (covering portion) [0165] **54** side face portion (pillar portion) [0166] **55** opening [0167] **90** seat [0168] **92** scat cushion [0169] **94** seatback

Claims

1. A seat switch device, comprising: a base portion; an operation knob that operates a seat and that is provided at a device front side with respect to the base portion; and a guard member that is provided at the device front side with respect to the base portion and that guards the operation knob.
2. The seat switch device of claim 1, wherein an end at a device front side of the guard member is positioned further toward the device front side than an end at a device front side of the operation knob.
3. The seat switch device of claim 1, wherein the guard member includes a covering portion that covers the operation knob from the device front side.
4. The seat switch device of claim 1, wherein: the guard member includes an outer face portion configuring an outer face of the guard member; and the outer face portion includes a top face portion configuring a top face facing toward the device front side, with the top face portion covering the operation knob from the device front side, and a side face portion configuring a side face facing in a direction perpendicular to a device front direction.
5. The seat switch device of claim 4, wherein the side face portion includes an opening exposing a portion of the operation knob.
6. The seat switch device of claim 1, wherein the guard member includes: a covering portion that covers the operation knob from the device front side; and two pillar portions that support the covering portion from a device rear side, with the two pillar portions respectively disposed on each side of the operation knob.
7. The seat switch device of claim 6, wherein: the seat switch device includes a guard projection portion that projects from the base portion toward the device front side; and the pillar portions include a side face portion configuring a side face facing in a direction perpendicular to a device front direction, and a fitting portion that fits together with the guard projection portion.
8. The seat switch device of claim 1, wherein: the seat switch device is arranged with a device front side facing toward a seat width direction outer side; the operation knob operates a seatback; the guard member has a shape with a length direction substantially along a seat height direction; the guard member includes side face portions configuring side faces of the guard member; and among the side faces, a rear side face facing toward a seat rear side extends in a direction inclined toward a seat rear on progression in a seat upward direction.

9. The seat switch device of claim 1, wherein: the seat switch device is arranged with a device front side facing toward a seat width direction outer side; the operation knob operates a seat cushion; the guard member has a shape with a length direction along a seat front-rear direction; the guard member includes side face portions configuring side faces of the guard member; and among the side faces, a lower side face facing in a seat downward direction extends in the seat front-rear direction.

10. A seat comprising the seat switch device of claim 1.